2- Why we use IActionResult not ActionResult?

IActionResult is an interface in ASP.NET Core MVC that allows controller actions to return different types of responses such as View, Json, NotFound, or Redirect. It provides flexibility and loose coupling compared to returning a specific result type.

3- What HTTPContext request and response consist of?

HttpContext represents the entire HTTP transaction.
It includes Request (method, headers, URL, body) and Response (status code, headers, body), allowing the server to read incoming data and send a response to the client.

4- Difference between HTTP and HTTPS?

HTTP transfers data in plain text, while HTTPS encrypts the communication using SSL/TLS, ensuring confidentiality, integrity, and authentication between the client and the server.

5- Segments and Fragments in URL?

URL segments are parts of the path used for routing (e.g., /products/electronics).
Fragments (#section) are client-side references used by the browser to navigate to a specific section of the page.

6- Builder Pattern vs Dependency Injection Real life example?

The Builder pattern constructs complex objects step by step.
Dependency Injection provides dependencies to a class from the outside, improving modularity, testability, and maintainability.

7- Web Pages (Razor) vs MVC?

Razor Pages follow a page-focused architecture suitable for simple applications, while MVC separates concerns into Model, View, and Controller, making it more suitable for large and complex systems.

8- Content-Type in Response?

The Content-Type header specifies the format of the response body (e.g., HTML, JSON, image). It allows the client to correctly interpret and process the returned data.

9- Minification, Bundling, Webpack, Lazy Loading?

Minification reduces file size by removing unnecessary characters.
Bundling combines multiple files into one.
Webpack automates bundling and optimization.
Lazy loading loads resources only when needed, improving performance and reducing network usage.

Bonus

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1 - File Based URL?

File-Based URL is a URL structure where the web address directly corresponds to a physical file stored on the web server.

2 - Clean URL?

A **Clean URL** is a simple, readable, and user-friendly web address that does not contain file names or complex query strings.

3 - Controller Factory?

Controller Factory is a component in ASP.NET MVC responsible for creating controller instances to handle incoming HTTP requests.

When a user sends a request, the routing system determines which controller should handle it. Then the Controller Factory creates an instance of that controller and returns it to the MVC framework, which executes the requested action method.

Controller factories can also be customized to control how controllers are created, support dependency injection, and integrate with IoC containers.

4 - N-Tier?

N-Tier Architecture is a software architecture that divides an application into multiple layers or tiers, where each tier has a specific responsibility.

The most common structure is the 3-Tier Architecture, which includes the Presentation Layer (user interface), Business Logic Layer (application rules and processing), and Data Access Layer (database operations).

This architecture improves code organization, maintainability, scalability, and allows different parts of the system to be developed and managed independently.

5- Clean Architecture ?

Clean Architecture is a software design approach that organizes an application into layers with a strong focus on separating business rules from external frameworks and technologies.

The architecture consists of four main layers: Entities, Use Cases, Interface Adapters, and Frameworks & Drivers. The central idea is the Dependency Rule, where dependencies always point inward toward the core business logic.

This approach improves maintainability, testability, and flexibility, allowing the system to evolve without being tightly coupled to specific frameworks or databases.